A cheap exact pre-test for GetDecics' field-discriminant test

Proposal, with proofs

Passiflora

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Abstract

In a Passiflora GPU task most of the CPU time now goes to one step of the original GetDecics: PARI's `discf`, the field discriminant of every irreducible polynomial that passes the polynomial-discriminant test. On the test work units of the sf5 and sf7 searches thousands of polynomials reach it per unit, and all but a handful fail. This document proposes a test that runs before `discf` and proves most of these failures with arithmetic modulo a few primes (Dedekind's criterion at primes dividing the polynomial discriminant). It proves that the test rejects only polynomials that `discf` would reject, so the program's output and its counters stay exactly the same, and measures the test on the 55,181 polynomials that reach `discf` in three test work units: it rejects 69 to 83 per cent of the failures at about 0.06 ms per polynomial, against 0.75 to 0.86 ms for `nfdisc`, with no wrong rejection.

The status of each statement is marked as in the whitepaper: [proved] (proved here), [standard result, quoted] (a textbook result, with a reference), [checked numerically] (measured; the method is stated). Section 2 fixes the notation, Section 3 gives the rejection rule, Section 4 proves Dedekind's criterion in the direction used, Section 6 states the pre-test and proves it exact, Sections 7 and 8 say what it leaves to `discf` and what it saves, and Section 9 is the implementation plan.

1 The step and what it costs

After the polynomial-discriminant test (`polDiscTest_cpu`, the step the whitepaper's pre-filter accelerates) GetDecics finishes every polynomial f that passed it with upstream's code, unchanged by Passiflora (`TgtMartinet.cpp`; Passiflora's GPU path runs the same lines in `nfFinishPolys`):

```
if(isirreducible(f_L)) {
  (*irrCount)++;
  GEN fldDisc = discf(f_L);       // upstream: "a computationally expensive task"
  for(int k=1;k<=numPrimes;++k)   // divide out the primes of S
    fldDisc = gdiv(fldDisc,gpowgs((GEN)Sg[k],ggval(fldDisc,(GEN)Sg[k])));
  if(gcmp1(gabs(fldDisc,DEFAULTPREC))) { (*fdCount)++;  /* output f */ }
}
```

So f is a result exactly when it is irreducible and its field discriminant has no prime factor outside the set S of the work unit. Profiling Passiflora's CUDA program on a GTX 1650 (one task, Windows, the CPU idle otherwise) gave [checked numerically]:

work unit	task	GPU kernels	`discf`	calls	results
`sf7_DS-12x12_Grp811471of1000000`	26.5 s	13.8 s	10.3 s	12,895	6
`sf5_DS-16x12_Grp64300of1600000`	38.5 s	13.5 s	22.0 s	21,274	2

While the GPU computes, the CPU sleeps; of the task's CPU time, `discf` took 84 and 89 per cent. The stock CPU program makes the same calls, but there they are lost in a run of over an hour. Of the polynomials that reach `discf`, 99.9 per cent fail.

2 Setting and notation

Throughout, $f \in \mathbb{Z}[x]$ is monic and irreducible over $\mathbb{Q}$ of degree n (in GetDecics $n = 10$), α is a root of f, $K = \mathbb{Q}(\alpha)$, $\mathcal{O}_K$ is the ring of integers of K, $R = \mathbb{Z}[\alpha] \subseteq \mathcal{O}_K$, d_K is the discriminant of K and $\operatorname{disc} f = \prod_{i<j}(\alpha_i - \alpha_j)^2$ the discriminant of f (the α_i the roots of f in $\mathbb{C}$). Both R and $\mathcal{O}_K$ are free $\mathbb{Z}$-modules of rank n, so the *index* $\operatorname{ind}(f) = [\mathcal{O}_K : R]$ is finite. S is a finite set of primes. For a prime p, a bar denotes reduction modulo p, v_p the p-adic valuation.

Definition 1. *f passes the field-discriminant test* for S if every prime dividing d_K lies in S.

This is the test of the code above: dividing $d_K \neq 0$ by $p^{v_p(d_K)}$ for every $p \in S$ leaves ± 1 exactly when no other prime divides d_K.

3 The rejection rule

Lemma 2 (the index formula)**.** $\operatorname{disc} f = \operatorname{ind}(f)^2 \, d_K$. *[standard result, quoted] [Neu99, Ch. I, Prop. 2.12]*

Proof. Let $\omega_1, \dots, \omega_n$ be a $\mathbb{Z}$-basis of $\mathcal{O}_K$ and write $\alpha^{j-1} = \sum_k M_{jk}\omega_k$ with $M \in \mathbb{Z}^{n\times n}$. For the embeddings $\sigma_1, \dots, \sigma_n$ of K, $(\sigma_l(\alpha^{j-1})) = M\,(\sigma_l(\omega_k))$, and the discriminant of a basis is the square of the determinant of this matrix, so $\operatorname{disc}(1, \alpha, \dots, \alpha^{n-1}) = (\det M)^2 d_K$. The left side is $\det(\alpha_l^{j-1})^2 = \prod_{i<j}(\alpha_i - \alpha_j)^2 = \operatorname{disc} f$ (Vandermonde), and $|\det M| = [\mathcal{O}_K : R]$ (the index of a full sublattice is the absolute value of the determinant of its basis in a basis of the lattice). □

Proposition 3 (the rejection rule)**.** *Let $p \notin S$ be a prime with $p \mid \operatorname{disc} f$ and $p \nmid \operatorname{ind}(f)$. Then $p \mid d_K$, and f fails the field-discriminant test for S. [proved]*

Proof. By Lemma 2, $v_p(d_K) = v_p(\operatorname{disc} f) - 2v_p(\operatorname{ind} f) = v_p(\operatorname{disc} f) \geq 1$, and $p \notin S$. □

Remark 4 (why the whitepaper's filter cannot help here)**.** Lemma 2 also gives $v_p(d_K) \equiv v_p(\operatorname{disc} f) \pmod 2$. The polynomial-discriminant test asks whether $|\operatorname{disc} f| = \big(\prod_{p\in S} p^{a_p}\big) m^2$ (the whitepaper's definition of *passes*, and its Proposition "What the exact test computes" for the code): whether $v_p(\operatorname{disc} f)$ is even for every $p \notin S$. So it is exactly the parity condition, and the whitepaper's filter is a cheap test of that condition (a Legendre symbol -1 modulo a filter prime proves an odd valuation somewhere). A polynomial that passes has $\operatorname{disc} f = \pm \prod_{p\in S} p^{a_p} \cdot m^2$; at every filter prime q ($q \equiv 1 \pmod 4$, every $p \in S$ a square modulo q) its Legendre symbol is therefore 0 or 1, and the filter can reject none of them. What remains is whether $\operatorname{ind}(f)$ absorbs all of m: by Lemma 2 f passes the field-discriminant test if and only if $v_p(\operatorname{ind} f) = v_p(m)$ for every prime $p \mid m$, $p \notin S$. That is a question about the ring $\mathbb{Z}[\alpha]$ at the primes dividing m, which residues of $\operatorname{disc} f$ do not see.

4 Dedekind's criterion

Fix a prime p and factor $\bar f = \prod_{i=1}^{r} \bar\varphi_i^{e_i}$ in $\mathbb{F}_p[x]$, with distinct monic irreducible $\bar\varphi_i$ and $e_i \geq 1$. Choose monic $\varphi_i \in \mathbb{Z}[x]$ reducing to $\bar\varphi_i$, and put

$$G = \prod_{i=1}^{r} \varphi_i, \qquad H = \prod_{i=1}^{r} \varphi_i^{e_i - 1}, \qquad F = \frac{GH - f}{p}.$$

Since $\bar G \bar H = \prod \bar\varphi_i^{e_i} = \bar f$, the polynomial $GH - f$ has coefficients divisible by p, so $F \in \mathbb{Z}[x]$.

Theorem 5 (Dedekind's criterion, the direction used)**.** *If $\gcd(\bar F, \bar G, \bar H) = 1$ in $\mathbb{F}_p[x]$, then $p \nmid \operatorname{ind}(f)$. [proved], from the standard facts quoted in Lemmas 6 and 7.*

The converse also holds [Ded78, Coh93] (Theorem 6.1.4 there), so the criterion decides whether $p \mid \operatorname{ind}(f)$, and in particular its outcome does not depend on the choice of the lifts φ_i; only the direction above is needed for exactness, and its proof below works for any choice. The proof takes three lemmas.

Lemma 6. *The map $\mathbb{Z}[x] \to R$, $x \mapsto \alpha$, is onto with kernel $f\mathbb{Z}[x]$, so $R/pR \cong \mathbb{F}_p[x]/(\bar f)$. The maximal ideals of R that contain p are $\mathfrak{m}_i = pR + \varphi_i(\alpha)R$ for $i = 1, \dots, r$, pairwise distinct, and for $v \in \mathbb{Z}[x]$: $v(\alpha) \in \mathfrak{m}_i \iff \bar\varphi_i \mid \bar v$. [proved]*

Proof. The map is onto by definition of R. If $g(\alpha) = 0$ for $g \in \mathbb{Z}[x]$, then $f \mid g$ in $\mathbb{Q}[x]$ because f is the minimal polynomial of α; division of g by the monic f in $\mathbb{Z}[x]$ has a remainder of degree $< n$ vanishing at α, hence 0, so $f \mid g$ in $\mathbb{Z}[x]$. Therefore $R/pR \cong \mathbb{Z}[x]/(p, f) \cong \mathbb{F}_p[x]/(\bar{f})$. The ideals of $\mathbb{F}_p[x]$ containing $\bar{f}$ are $(\bar{d})$ for the monic divisors $\bar{d} \mid \bar{f}$, and those that are maximal are $(\bar{\varphi}_i)$. Their preimages in R are the $\mathfrak{m}_i$, distinct because the $\bar{\varphi}_i$ are, and every maximal ideal of R containing p contains pR, so it is one of them. Finally $v(\alpha) \in \mathfrak{m}_i$ if and only if $\bar{v}$ lies in $(\bar{\varphi}_i)$ modulo $\bar{f}$, that is $\bar{\varphi}_i \mid \bar{v}$, since $\bar{\varphi}_i \mid \bar{f}$. □

Lemma 7. *$p \nmid \operatorname{ind}(f)$ if and only if, for every i, the maximal ideal $\mathfrak{m}_i R_{\mathfrak{m}_i}$ of the local ring $R_{\mathfrak{m}_i}$ is principal. [proved], from [AM69, Prop. 5.12, 5.13, Cor. 5.8, Prop. 9.2]*

Proof. Let $T = \mathbb{Z} \setminus p\mathbb{Z}$. The group $Q = \mathcal{O}_K/R$ is finite of order $\operatorname{ind}(f)$, and localization is exact, so $T^{-1}Q = T^{-1}\mathcal{O}_K/T^{-1}R$; for a finite abelian group $T^{-1}Q$ is its p-primary part. Hence $p \nmid \operatorname{ind}(f)$ if and only if $T^{-1}R = T^{-1}\mathcal{O}_K$. Since $\mathcal{O}_K$ is the integral closure of R in K, $T^{-1}\mathcal{O}_K$ is the integral closure of $T^{-1}R$ [AM69, Prop. 5.12], so $p \nmid \operatorname{ind}(f)$ if and only if $T^{-1}R$ is integrally closed, which holds if and only if its localization at each maximal ideal is integrally closed [AM69, Prop. 5.13].

R is a noetherian domain (a finitely generated $\mathbb{Z}$-module inside K), integral over $\mathbb{Z}$. A nonzero prime $\mathfrak{p}$ of R contracts to a nonzero prime $q\mathbb{Z}$: for $0 \neq x \in \mathfrak{p}$ the minimal polynomial $x^d + c_{d-1}x^{d-1} + \cdots + c_0$ of x over $\mathbb{Q}$ has integer coefficients (x is integral) and $c_0 \neq 0$, and $c_0 = -x(x^{d-1} + \cdots + c_1) \in xR \subseteq \mathfrak{p}$. Then $\mathfrak{p}$ is maximal [AM69, Cor. 5.8]. The prime ideals of $T^{-1}R$ correspond to the primes of R disjoint from T: the zero ideal and the nonzero primes contracting to $p\mathbb{Z}$, which by Lemma 6 are the $\mathfrak{m}_i$ ($r \geq 1$, so $T^{-1}R$ is not a field). So the maximal ideals of $T^{-1}R$ are the $T^{-1}\mathfrak{m}_i$, and $(T^{-1}R)_{T^{-1}\mathfrak{m}_i} = R_{\mathfrak{m}_i}$. Each $R_{\mathfrak{m}_i}$ is a noetherian local domain whose primes are 0 and its maximal ideal, so of dimension one, and for such a ring being integrally closed is equivalent to its maximal ideal being principal [AM69, Prop. 9.2]. □

Lemma 8. $\gcd(\bar{G}, \bar{H}) = \prod_{e_i \geq 2} \bar{\varphi}_i$, *so* $\gcd(\bar{F}, \bar{G}, \bar{H}) = 1$ *if and only if no $\bar{\varphi}_i$ with $e_i \geq 2$ divides $\bar{F}$. [proved]*

Proof. $\bar{G}$ is the product of the distinct irreducibles $\bar{\varphi}_i$, each to the first power, and $\bar{\varphi}_i \mid \bar{H}$ exactly when $e_i \geq 2$. □

Proof of Theorem 5. From $f(\alpha) = 0$ and the definition of F,

$$G(\alpha)\, H(\alpha) = p\, F(\alpha). \tag{1}$$

Fix i and let $A = R_{\mathfrak{m}_i}$. By Lemma 6, $v(\alpha)$ is a unit of A for every $v \in \mathbb{Z}[x]$ with $\bar{\varphi}_i \nmid \bar{v}$. Write $G = \varphi_i G_i$ with $G_i = \prod_{j \neq i} \varphi_j$; the $\bar{\varphi}_j$ are distinct irreducibles, so $\bar{\varphi}_i \nmid \bar{G}_i$ and $G_i(\alpha)$ is a unit of A. By Lemma 7 it suffices to show that $\mathfrak{m}_i A = pA + \varphi_i(\alpha)A$ is principal.

Case $e_i = 1$. Then $\bar{\varphi}_i \nmid \bar{H}$, so $H(\alpha)$ is a unit of A, and (1) gives $\varphi_i(\alpha) = p\, F(\alpha) \big/ \big(G_i(\alpha) H(\alpha)\big) \in pA$. So $\mathfrak{m}_i A = pA$.

Case $e_i \geq 2$. By the hypothesis and Lemma 8, $\bar{\varphi}_i \nmid \bar{F}$, so $F(\alpha)$ is a unit of A, and (1) gives $p = \varphi_i(\alpha)\, G_i(\alpha) H(\alpha) / F(\alpha) \in \varphi_i(\alpha)A$. So $\mathfrak{m}_i A = \varphi_i(\alpha)A$. □

5 Which primes to try

Lemma 9. *$p \mid \operatorname{disc} f$ if and only if $e_i \geq 2$ for some i. [proved]*

Proof. $\operatorname{disc} f = (-1)^{n(n-1)/2} \operatorname{Res}(f, f')$ is a polynomial with integer coefficients in the coefficients of the monic f, so $\overline{\operatorname{disc} f} = \operatorname{disc} \bar{f}$, the discriminant of the monic $\bar{f}$ of degree n over $\mathbb{F}_p$, which vanishes exactly when $\bar{f}$ has a repeated root in an algebraic closure. Over the perfect field $\mathbb{F}_p$ each $\bar{\varphi}_i$ has simple roots and distinct $\bar{\varphi}_i$ have no common root, so a repeated root exists exactly when some $e_i \geq 2$. □

So at a prime $p \nmid \operatorname{disc} f$ all $e_i = 1$, the hypothesis of Theorem 5 holds trivially, and it says only what Lemma 2 already says; and Proposition 3 needs $p \mid \operatorname{disc} f$. The primes worth trying are the primes $p \notin S$ dividing $\operatorname{disc} f$, that is (Remark 4) the primes of m. Small ones are found by trial division. Once they are divided out, what is left of $|\operatorname{disc} f|$ without the primes of S is often a prime power q^k (Section 8), and then q is one more prime to try. Theorem 5 needs q to be prime, not merely probably prime: a primality *proof* is required (PARI's `isprime` gives one; for $q < 2^{64}$, which covers every case measured here, the BPSW test that it uses is known to be correct).

6 The pre-test and its exactness

Definition 10 (the pre-test)**.** For f monic and irreducible, S, and a bound B:

1. Compute $D = \operatorname{disc} f$ exactly, and $N = |D|$ with every prime of S divided out.
2. For each prime $p < B$, $p \notin S$, in increasing order: if $p \mid N$, then if $\textsc{Dedekind}(f, p)$ holds, *reject*; otherwise divide every factor p out of N.
3. If $N > 1$ and $N = q^k$ with q proved prime, and $\textsc{Dedekind}(f, q)$ holds, *reject*.
4. Otherwise the pre-test is *undecided*, and the program computes `discf` as now.

$\textsc{Dedekind}(f, p)$: factor $\bar{f}$ over $\mathbb{F}_p$, form G, H and F as in Section 4, and test $\gcd(\bar{F}, \bar{G}, \bar{H}) = 1$.

Theorem 11 (exactness)**.** *If the pre-test rejects f, then f fails the field-discriminant test for S.* [proved]

Proof. A rejection happens at a prime p ($p < B$ in step 2, $p = q$ in step 3) with $p \notin S$ (the primes of S were divided out of N) and $p \mid N \mid D = \operatorname{disc} f$, at which $\textsc{Dedekind}(f, p)$ holds. By Theorem 5, $p \nmid \operatorname{ind}(f)$, and by Proposition 3, f fails. □

Corollary 12 (the output is unchanged)**.** *Run the pre-test after* `isirreducible` *and after* `irrCount` *is incremented, and compute* `discf` *only when it is undecided. Then the counters* `pdCount`*,* `irrCount`*,* `fdCount` *and the output file, polynomial by polynomial and in order, are those of the program without the pre-test.* [proved]

Proof. The pre-test changes nothing before the test of d_K. A polynomial it rejects fails that test by Theorem 11, so without the pre-test it would neither increment `fdCount` nor be written; every other polynomial goes through `discf` and the same test as before. □

The corollary assumes that the arithmetic is exact: the discriminant and the divisions in $\mathbb{Z}$, the factorization over $\mathbb{F}_p$ (Berlekamp's or Cantor–Zassenhaus' algorithm; the latter is randomized in its running time only, its output is always a correct factorization) and the primality proof. PARI provides all of them, and the program already trusts PARI for `discf` itself.

7 What the pre-test leaves to `discf`

A polynomial that passes the field-discriminant test is always left to `discf` (contrapositive of Theorem 11). A polynomial that fails is left to it when, at every prime $p \notin S$ dividing d_K, either

(a) p divides the index: by the converse of Theorem 5 the criterion then fails at p, although $v_p(\operatorname{ind} f) < v_p(m)$; or

(b) $p \geq B$ and the part of m left after trial division is not a prime power (p is never tried).

Two extensions would reach some of these. For (a), the theorem of Ore [Ore28] and its generalization by the Montes algorithm [GMN12] compute $v_p(\operatorname{ind} f)$ from Newton polygons of f when f is p-regular, still with arithmetic modulo powers of p. PARI's order maximal at a given prime (`nfdisc([f,[p]])`) decides (a) completely, but in the measurements below it cost more than `nfdisc` itself. For (b), a larger bound B helps (Section 8); factoring m completely is part of what makes `discf` expensive.

8 Measurements

Method [checked numerically]. An instrumented copy of Passiflora's CUDA program (not the repository's code) wrote every polynomial it passed to `discf`, on three test work units. PARI/GP 2.17.3 then computed for each the true answer with `nfdisc`, ran the pre-test (the functions of Appendix A) for several bounds, and checked every rejection against `nfdisc`: d_K must contain the prime the pre-test named. Times are GP's on an i5-12500H; the program links PARI 2.8, whose `discf` took 0.80 and 1.03 ms per polynomial on the first two units.

work unit (Grp)	sf7, 811471	sf5, 64300	sf5, 82890
polynomials reaching `discf`	12,895	21,274	21,012
failing the field-discriminant test	12,889	21,272	21,005
smallest bad prime below 1000	10,107	17,442	17,234
smallest bad prime above 1000	2,782	3,830	3,771
rest of m after the primes < 1000: 1	830	193	43
a prime power	6,998	2,130	965
composite	5,067	18,951	20,004
failures rejected: steps 1–2, $B = 100$	6,304 (48.9%)	11,218 (52.7%)	11,116 (52.9%)
steps 1–2, $B = 1000$	8,383 (65.0%)	14,500 (68.2%)	14,413 (68.6%)
steps 1–2, $B = 10^4$	9,493 (73.7%)	16,260 (76.4%)	16,124 (76.8%)
steps 1–3, $B = 100$	7,556 (58.6%)	11,298 (53.1%)	11,126 (53.0%)
steps 1–3, $B = 1000$	10,719 (83.2%)	15,005 (70.5%)	14,530 (69.2%)
wrong rejections, every variant	0	0	0
failures left by case (a), bad prime < 1000	1,724	2,942	2,821
time per polynomial: steps 1–3, $B = 1000$	0.074 ms	0.057 ms	0.058 ms
steps 1–2, $B = 10^4$	0.103 ms	0.098 ms	0.089 ms
`nfdisc`	0.747 ms	0.861 ms	0.780 ms

The work units are `sf7_DS-12x12_Grp811471of1000000`, `sf5_DS-16x12_Grp64300of1600000` and `sf5_DS-16x12_Grp82890of1600000`; the shares are of the failures.

Every one of the 10,093 prime powers found in step 3 had $q < 2^{40}$, where `isprime` proves primality in a few microseconds. With steps 1–3 and $B = 1000$ the time for the polynomials that reach `discf` would drop, if the polynomials left over cost the average `nfdisc` time, from 9.6 to 2.6 s on the sf7 unit (−73%), from 18.3 to 6.6 s (−64%) and from 16.4 to 6.3 s (−62%) on the sf5 units. In the profiled GPU task of the sf7 unit this would take its CPU time from about 12 s to about 4 to 5 s. Steps 1–3 with $B = 10^4$ were not measured; the table suggests they would combine the gains of both.

9 Implementation plan

1. Add the pre-test at both places that call `discf`: upstream's loop in `TgtMartinet.cpp` (through `patches/passiflora.patch`) and `nfFinishPolys`, after `irrCount` is incremented (Corollary 12).
2. Use PARI's library functions, which PARI 2.8 has: `ZX_disc` for D, `Z_lvalrem`/`Z_pvalrem` for the divisions, `Z_isanypower` and `isprime` for step 3, and factorization over $\mathbb{F}_p$ (`FpX_factor`, or the word-size `Flx` functions, since every p here is below 2^{40}) with `FpX_gcd` for Dedekind. Choose B from measurements of the C code (1000 to 10^4).
3. With `NF_VERIFY=1`, also compute `discf` for every polynomial the pre-test rejects and stop if it passes, so that a fault in the implementation shows up.
4. Tests: all 31 upstream work units against their `.truth` files on the CPU and the GPU; a check that runs the pre-test on every polynomial that reaches `discf` in all 31 units and compares with `discf` (no wrong rejection allowed); the CPU time of GPU tasks before and after, on an idle machine.
5. Documents: a section of the whitepaper with Sections 3–6 of this proposal; notes in `TECHNICAL.md` and in `CHANGES-FROM-GETDECICS.md`.

A The GP functions used in the measurements

```
S = [2, 5];
spart(d) = { my(r = d); for (i = 1, #S, r /= S[i]^valuation(r, S[i])); r };
dedekind_ok(f, p) = {      \\ gcd(F, G, H) = 1 mod p, Theorem 4
  my(fa = factormod(f, p), G = 1, H = 1, F, T);
  for (i = 1, #fa~, my(g = lift(fa[i, 1])); G *= g;
                    if (fa[i, 2] > 1, H *= g^(fa[i, 2] - 1)));
  F = (G * H - f) / p;
  if (denominator(content(F)) != 1, error("G*H != f mod p"));
  T = gcd(gcd(F * Mod(1, p), G * Mod(1, p)), H * Mod(1, p));
```

```
  poldegree(T) == 0
};
pretest(f, D, B) = {       \\ steps 1-3; returns the prime of a rejection, or 0
  my(c = abs(spart(D)), q);
  forprime (p = 3, B - 1,
    if (setsearch(S, p) || c % p, next);
    if (dedekind_ok(f, p), return(p));
    while (c % p == 0, c /= p));
  if (c > 1, if (!isprime(c), ispower(c, , &q); if (q && isprime(q), c = q)));
  if (c > 1 && isprime(c) && dedekind_ok(f, c), return(c));
  0
};
```

The measurements of steps 1–2 tested $p \mid D$ directly. Those of steps 1–3 used the equivalent form on $m = \sqrt{N}$ (every N was a square, as the polynomial-discriminant test implies), with `ispseudoprime` in place of `isprime`. The code above, run as printed with $B = 1000$, rejects the same polynomials on all three units (10,719, 15,005 and 14,530), every rejection correct and at a prime that `isprime` proved.

References

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